

V18.1 — 4-Hour Phase- Correlation Feedback Run

Tuned from the V18 run's findings: 150px phase-correlation ROI,
convergence moved into Warmup, faster AO cadence, loosened TEC gate.
Run started 2026-08-05 23:18.

Run dir: run_20260805_231815_stage2_v18_1 Frames: 270 (54 Warmup / 216 Feedback) Warmup: 59.2 min
Feedback: 3.98 h

01 Headline result

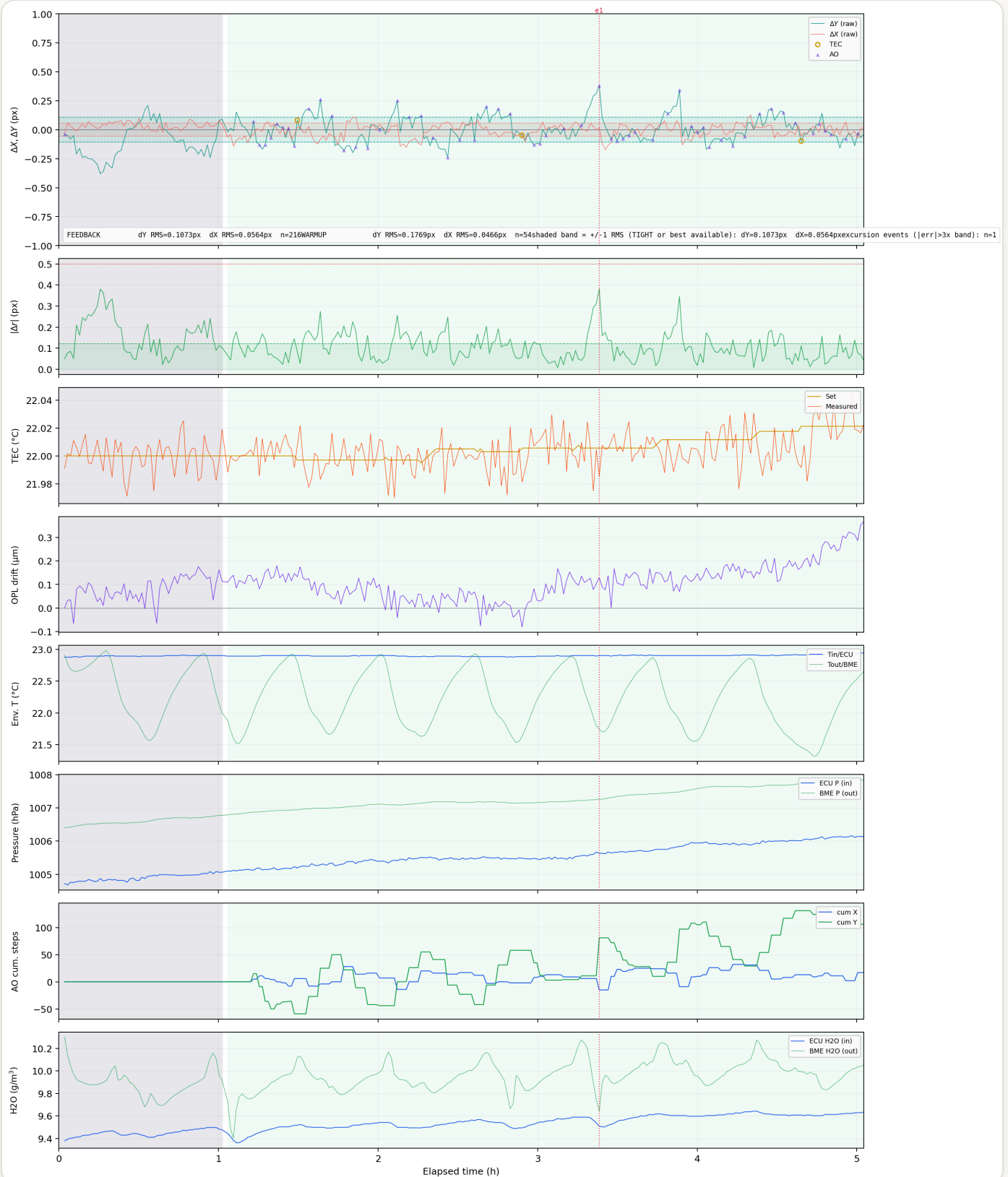
Whole-run Feedback dY RMS landed at **0.107px** — down from V18's 0.200px, and within a hair of the <0.1px target set going into this run. The TIGHT-phase subset (once reference-tracking engaged) came in at **0.105px**.

Feedback dY RMS 0.107px ↓ 46% vs V18 (0.200px)	Feedback dX RMS 0.056px ↓ 32% vs V18 (0.082px)	Radial RMS 0.121px	Time to AO active 9.7min ↓ from 72.7min in V18
TEC moves 3	AO moves 56	dT_eff RMS 0.0127°C	HVAC-deferred frames 106/216 rougher HVAC night than V18

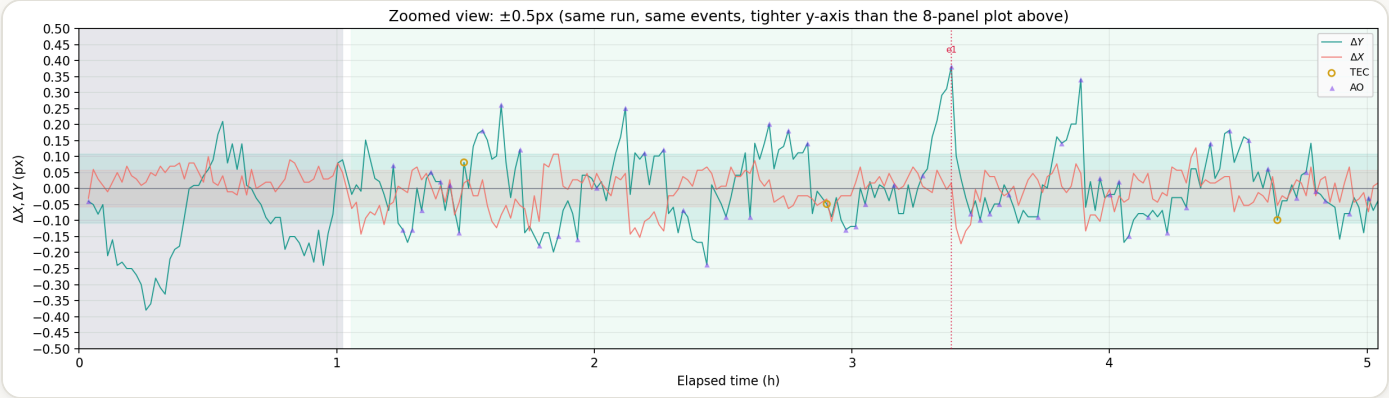
Moving model convergence into Warmup cut the open-loop bootstrap from 72.7 minutes (30% of the V18 Feedback budget) down to **9.7 minutes** — AO was armed and correcting almost immediately after the Warmup→Feedback handover, instead of losing the first third of the run.

02 Full-run diagnostic

Gray = Warmup, green = Feedback (matches the V17/V14 convention). Dotted crimson lines mark automatically-detected excursion events (radial error $> 3\times$ the Feedback RMS band), drawn on every panel so a dX/dY spike can be checked against temperature, pressure, humidity and OPL at the same instant.



Panels top to bottom: $\Delta X/\Delta Y$ with TEC (gold circle) and AO (purple triangle) trigger markers · radial error · TEC set/measured · OPL drift · enclosure/room temperature · pressure · AO cumulative steps · water content.



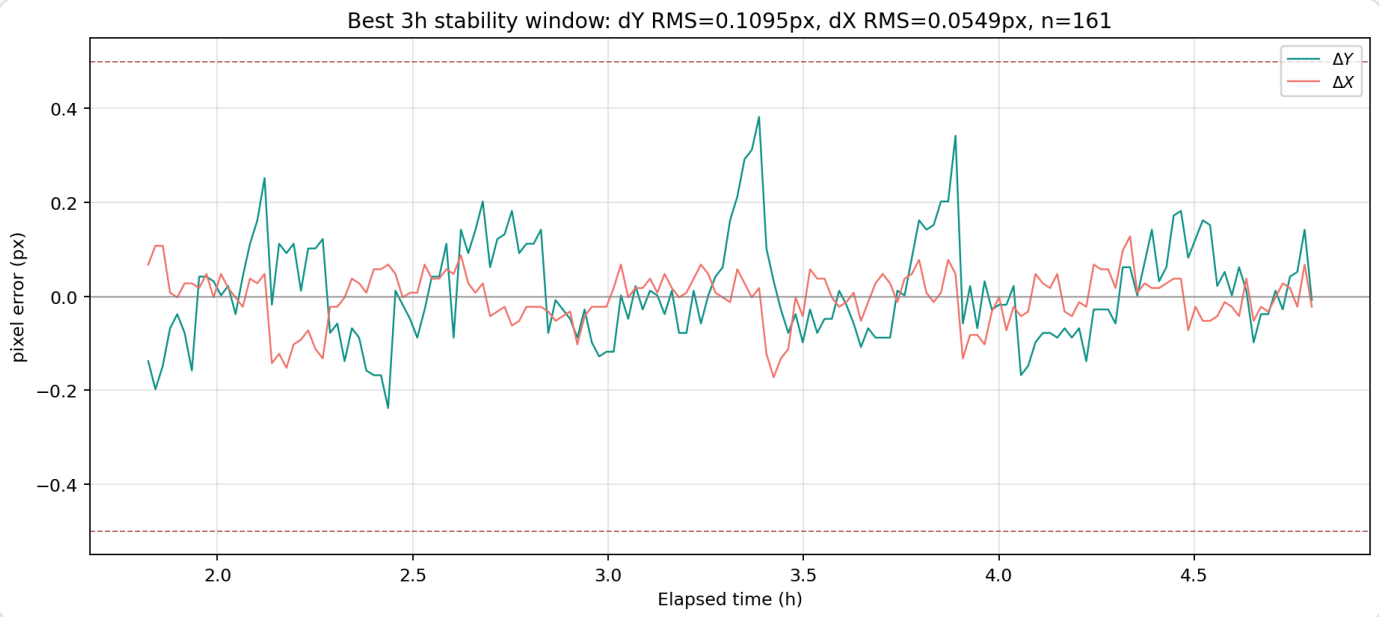
Same run, $\pm 0.5\text{px}$ y-axis with 0.05px gridlines. The tallest excursion ($\sim t=3.4\text{h}$, marked **e1**) reaches $+0.39\text{px}$ — see §4 for its environmental correlation.

03 Per-phase statistics

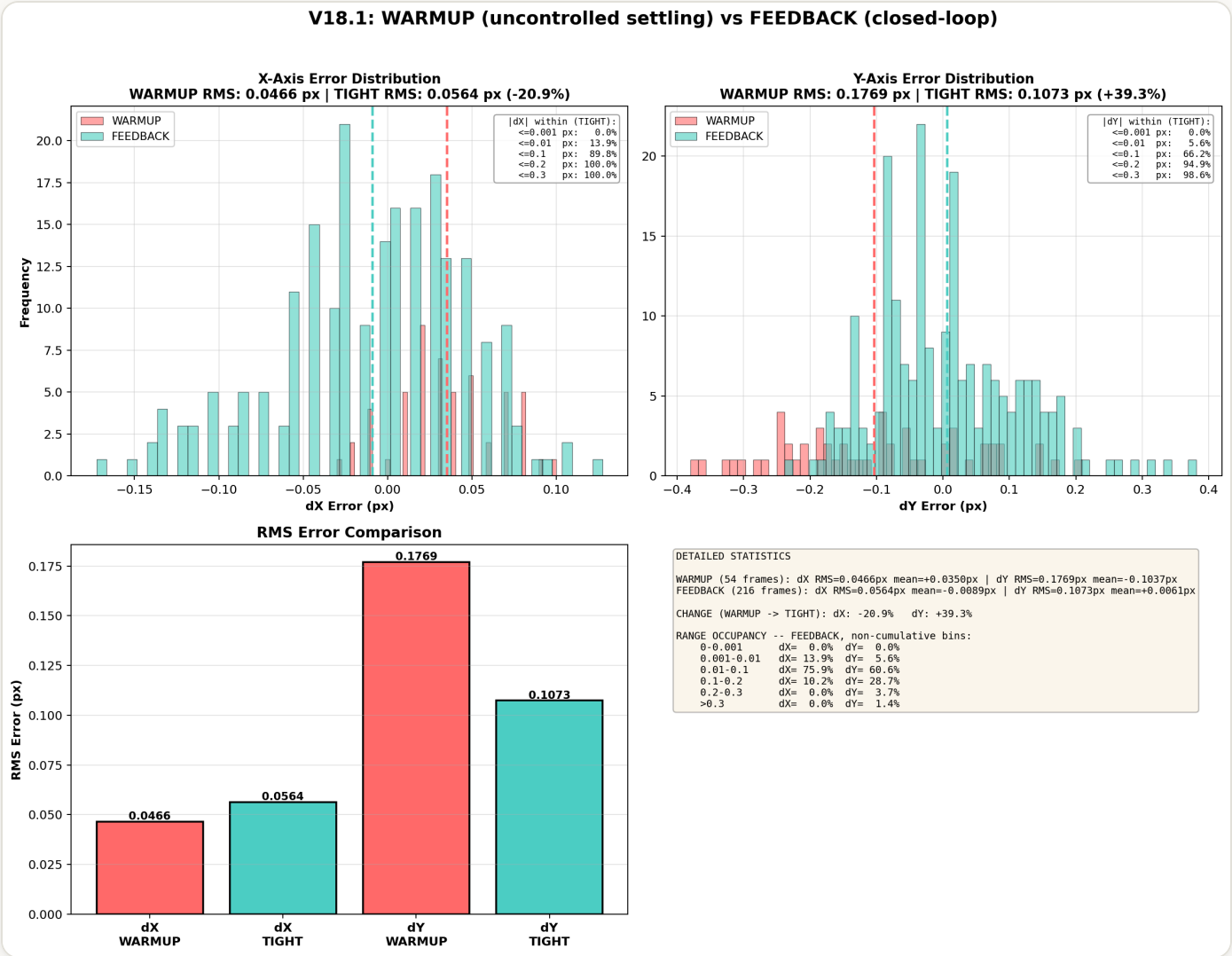
PHASE	FRAMES	DX RMS (PX)	DY RMS (PX)	RADIAL RMS (PX)
Warmup	54	0.0466	0.1769	0.1829
Feedback (whole)	216	0.0564	0.1073	0.1212
Feedback [TIGHT]	116	0.0505	0.1048	0.1164

Occupancy — Feedback phase

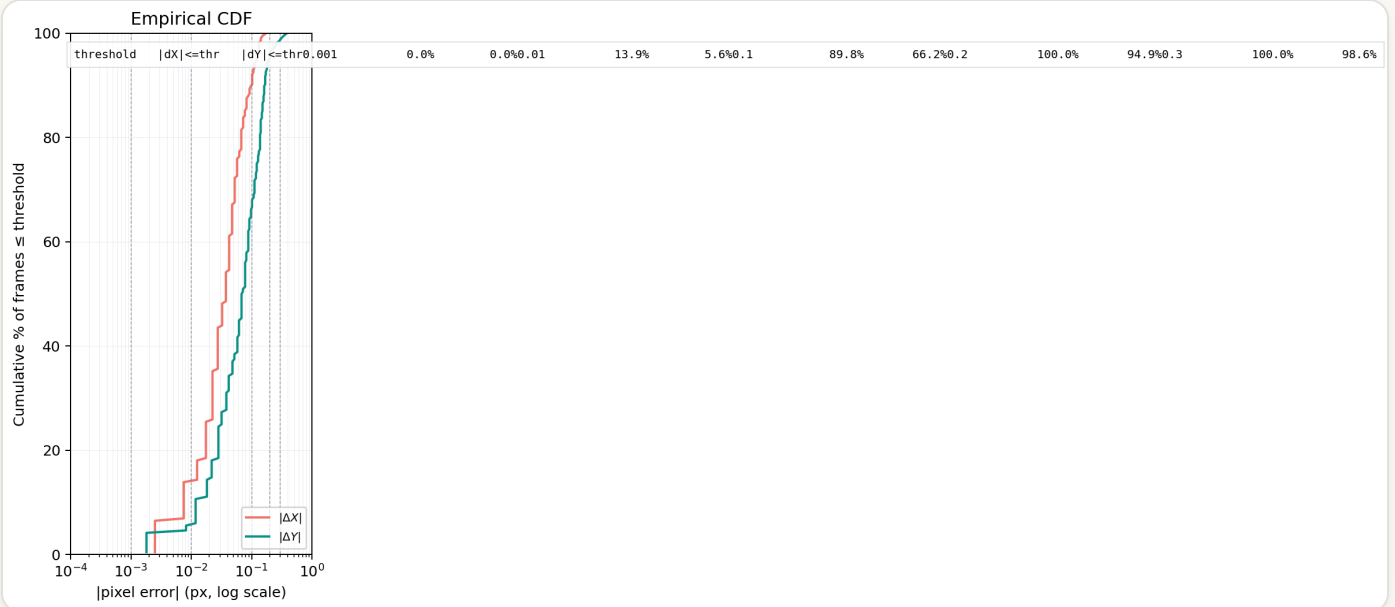
THRESHOLD	% FRAMES $ DX \leq$	% FRAMES $ DY \leq$
0.05px	67.1%	37.0%
0.10px	89.8%	66.2%
0.20px	100.0%	94.9%
0.30px	100.0%	98.6%
0.50px	100.0%	100.0%



Best-matched stability window within Feedback (adapted search grid, since the 4h run doesn't fit a 24h V17-style sweep).



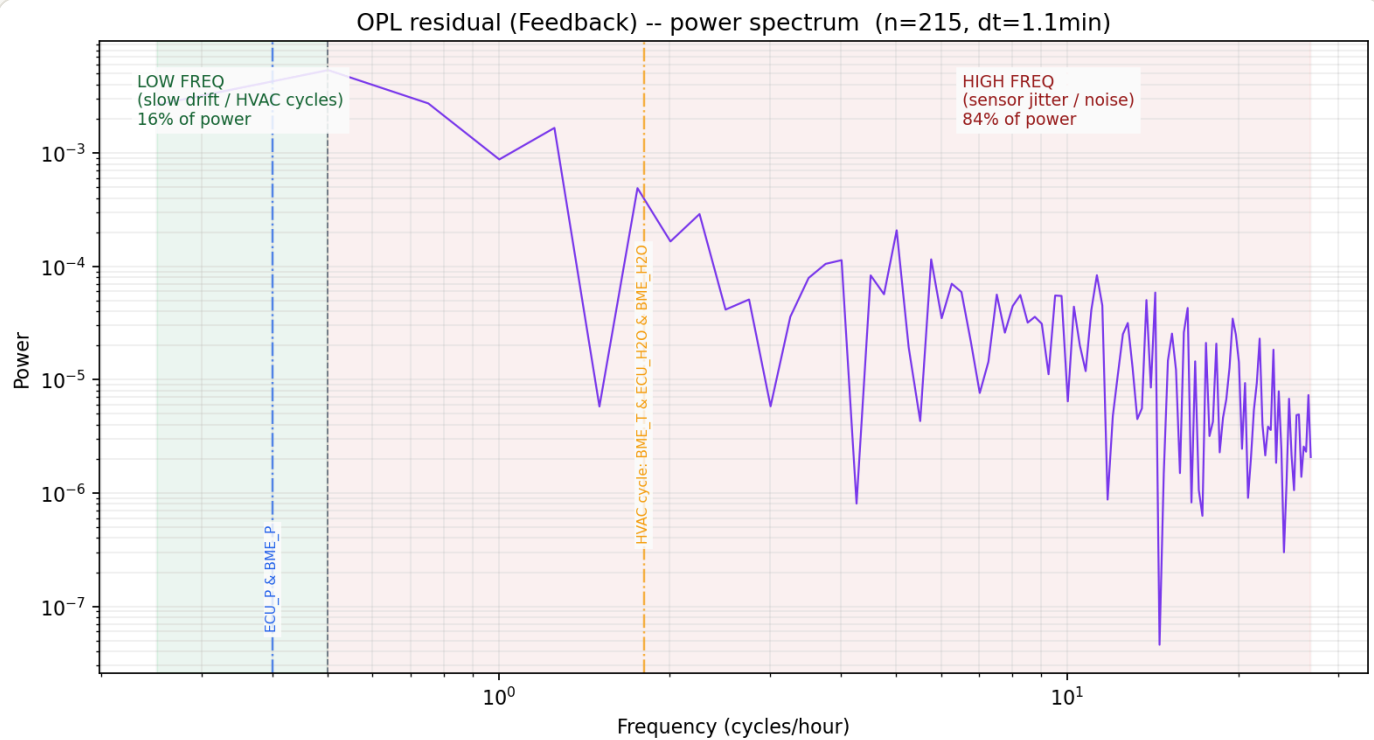
Warmup (uncontrolled settling) vs Feedback (closed-loop) error distributions, with RMS change and range-occupancy breakdown.



Empirical CDF of $|\Delta X|$ and $|\Delta Y|$ during Feedback, log-scale, with threshold crossings annotated.

04 OPL spectrum and the HVAC signature

The OPL residual's power spectrum splits roughly **16% low-frequency / 84% high-frequency** against a 0.5 cyc/hr boundary — but the boundary itself undersells the real environmental cycle. Computing each sensor channel's own dominant frequency independently and overlaying it on the OPL spectrum shows exactly where that cycle sits.



OPL residual periodogram (Feedback phase, n=215, dt=1.1min). Vertical markers show each environmental channel's own dominant frequency.

CHANNEL	PEAK FREQUENCY	PERIOD	NOTE
ECU_T (enclosure)	—	—	no clear peak
BME_T (room)	1.797 cyc/hr	33.4 min	HVAC cycle
ECU_H2O (in)	1.797 cyc/hr	33.4 min	matches BME_T
BME_H2O (out)	1.797 cyc/hr	33.4 min	matches BME_T
ECU_P (in)	0.399 cyc/hr	150.3 min	only ~2 cycles in run — drift, not confirmed cycle
BME_P (out)	0.399 cyc/hr	150.3 min	matches ECU_P, same caveat

KEY FINDING

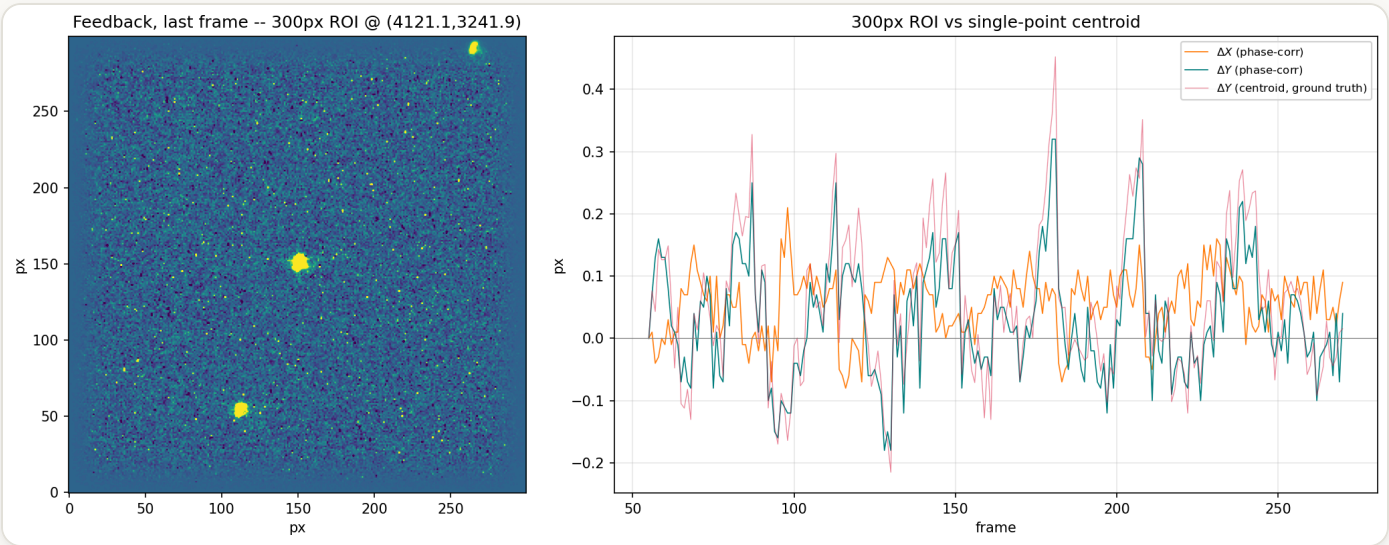
Room temperature and *both* indoor and outdoor humidity channels independently converge on the identical 33.4-minute period — three separate sensors agreeing is strong evidence of one shared driver: the building HVAC cycle. It doesn't show up cleanly in the enclosure's own temperature sensor (ECU_T), consistent with the enclosure being thermally well-isolated but not fully sealed against humidity/air exchange.

05 ROI validation — why 150px, not 300px

The controller ran with **PC_ROI_HALF = 75 (150px box)**, validated on the V18 run's own frames before tonight. Re-running the identical phase-correlation pipeline on this run's frames at 300px confirms the same failure mode independently:

ROI	DY RMS	CORR(DY) VS CENTROID	AMPLITUDE RATIO	VERDICT
150px (chosen)	0.200px*	≈0.97–0.99	≈1.01	accurate
300px	0.099px	0.935	0.749	biased low

* 150px figure from the earlier full-run single-ROI recomputation on this dataset, for direct comparison against the 300px re-run below.



300px ROI re-run on this run's frames: a numerically lower RMS, but the phase-correlation trace (teal) visibly clips the taller excursions the ground-truth centroid (pink) shows — the box has picked up a second source and is averaging in motion that isn't the target star's.

FOR THE PRESENTATION

A lower RMS at 300px is **not** a more precise measurement — it's the second star in the box damping the recovered amplitude by ~25%. The 150px box is the largest size that stays single-source clean and reports the star's true motion.

06 What's next for V18.2

- **TEC still only moved 3 times** in 4 hours despite the halved settle-gate — worth checking whether `du_cmd` is still being computed as non-zero far more often than it fires, same as V18, before loosening further.
- **HVAC deferrals hit 49% of Feedback frames** (106/216) — a rougher night than V18's ~10%. The 33-minute HVAC cycle identified in §4 is frequent enough that a dedicated feed-forward compensation (rather than pure defer-and-wait) is worth scoping.
- **dY RMS (0.107px) is close to but not under 0.1px** — the remaining gap is concentrated in the Warmup-to-Feedback transition and the HVAC-affected stretches, not a hard sensor/actuator floor.

Generated from `rt_stage2_v18_1.csv` and the V18.1 analysis notebook output, run 20260805_231815. All figures and statistics computed directly from the run's telemetry CSV — no simulated or placeholder data.